

A Stability Calculus for Local Learning Rules

Abstract

Algorithmic stability asks how much a learning rule changes when its training sample is perturbed. For local rules such as nearest neighbors, however, the phrase “change one sample point” hides several inequivalent operations: deleting an occurrence, replacing it by an adversarial labeled point, appending an occurrence, or deleting an occurrence and evaluating at that same occurrence as in leave-one-out validation. We develop a finite-sample stability calculus for deterministic k -nearest neighbors on finite metric spaces under explicit tie-breaking. The paper separates uniform perturbation relations from pointwise local phenomena: replace-one perturbations admit a clean delete-plus-add decomposition at the operation level, while leave-one-out stability need not control pointwise replace-one stability even for a two-point graph metric. We give an exact signed-margin characterization of when a deterministic k -NN prediction changes under a perturbation, visualize the resulting instability mechanisms, and provide reproducible finite-graph witness and certificate artifacts. Our odd- k extension currently remains computational evidence rather than theorem-level lifting. These finite worst-case separations complement, rather than contradict, the classical distributional consistency theory of nearest-neighbor rules.

1 Introduction

Algorithmic stability is often introduced through a single intuitive gesture: change one training example and ask whether the learner changes much. For global smooth learners this slogan is usually harmless. For local rules such as deterministic k -nearest neighbors, it is already underspecified. One can delete an occurrence, append a new occurrence, replace an occurrence by an adversarially chosen labeled point, or delete an occurrence and then evaluate the modified classifier exactly at that removed occurrence, as in leave-one-out (LOO) validation. These operations are related, but they are not identical. A substantial amount of avoidable confusion enters the discussion when they are treated as interchangeable.

This paper is about making those distinctions explicit in the smallest setting where they become mathematically sharp: deterministic k -NN on finite metric spaces with ordered samples, duplicate occurrences allowed, conflicting labels allowed, and tie-breaking frozen once and for all. The finite setting strips away distributional averaging and exposes the local combinatorics directly. In that regime we can ask a very concrete question: when does a single perturbation change the prediction, and what kinds of perturbations are provably stronger than others?

Figure 1 previews the perspective. Delete-one, add-one, replace-one, and LOO belong to the same family, but they move along different axes. Replace-one is strongest at the operation level because it can be factored into a deletion followed by an insertion; LOO is special because it evaluates at the removed occurrence itself. Once these axes are disentangled, two kinds of results emerge. First, there is a clean *uniform calculus*: replace-one control can be upper bounded by delete-one plus add-one control. Second, there are unexpectedly sharp *pointwise separations*: a sample can be perfectly LOO-stable while still being highly unstable under replace-one.

The smallest explicit witness already appears for deterministic 1-NN on a two-vertex graph metric. In that example, deleting either occurrence does not change the loss at the deleted occurrence, but replacing one occurrence by an adversarially labeled duplicate flips the prediction at a chosen query point. The witness is tiny enough to verify by hand, yet it already captures the main moral of the paper: LOO and replace-one are not merely different formalizations of the same local phenomenon.

Our contributions are therefore best read as a controlled bundle rather than a single theorem.

- (1) We formalize the perturbation vocabulary for deterministic finite k -NN, with explicit conventions for ordered samples, duplicate occurrences, conflicting labels, and tie-breaking.

Three perturbation notions differ by both operation and evaluation point

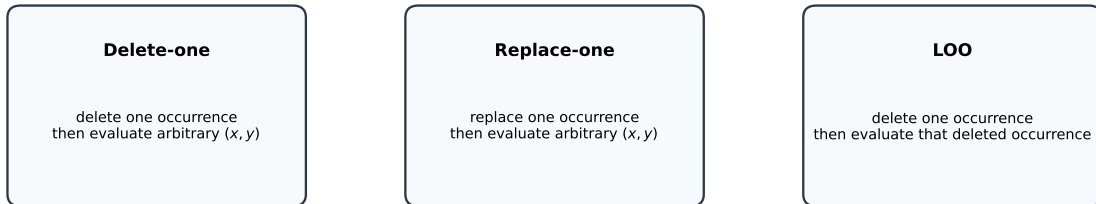


Figure 1: The paper separates perturbation operations by what changes in the sample and where the post-perturbation classifier is evaluated. This distinction is the organizing device for the rest of the draft.

- (2) We prove a uniform perturbation relation showing that replace-one can be decomposed into delete-one plus add-one at the loss level.
- (3) We derive an exact signed-margin criterion for when a deterministic k -NN prediction changes after a perturbation. This makes the instability mechanism transparent: the label vote inside the ordered top- k neighborhood must cross zero.
- (4) We give an explicit finite metric separation for deterministic 1-NN: pointwise LOO stability does not control pointwise replace-one stability. The witness itself is theorem-level; the current minimality claim remains a computational certificate.
- (5) We provide reproducible odd- k gadget candidates and publication assets that keep theorem statements, computational certificates, and conjectural lifting patterns cleanly separated.

The paper is intentionally modest about what it does *not* establish. We do not claim a new generalization bound. We do not claim that the current odd- k search outputs constitute a theorem. We do not claim that finite worst-case instability contradicts classical consistency results for nearest neighbors. Instead, we offer a definition-level and mechanism-level account of single-perturbation behavior that can live compatibly alongside the classical distributional theory.

The reader’s map is simple. [Section 2](#) positions the paper relative to classical algorithmic stability, deleted-estimate, and conformal prediction literatures. [Section 3](#) fixes the deterministic finite setting and notation. [Section 4](#) proves the uniform delete/add/replace relation. [Section 5](#) gives the exact deterministic k -NN perturbation criterion and the corresponding visual microscope for instability. [Section 6](#) states the minimal 1-NN separation result and its certificate boundary. [Section 7](#) summarizes what the current odd- k gadget search does and does not justify. [Section 8](#) explains why these results are compatible with consistency and prediction-calibration perspectives, and [Section 9](#) collects the main limitations and next steps.

2 Related Work

Classical algorithmic stability studies how a learning rule changes when the training sample is perturbed, primarily in order to control generalization under an i.i.d. sampling model. The canonical reference is Bousquet and Elisseeff, who formalize several stability notions around replace-one perturbations and fresh-test evaluation points [1]. Kutin and Niyogi expand this landscape with weaker and almost-everywhere variants [11], while Feldman and Vondrák and later Bousquet, Klochkov, and Zhivotovskiy sharpen high-probability and rate-sensitive uniform stability bounds [8, 2]. Our paper does not compete with this line on sharper generalization bounds. Instead, we use it as the baseline vocabulary against which a more microscopic perturbation calculus can be compared.

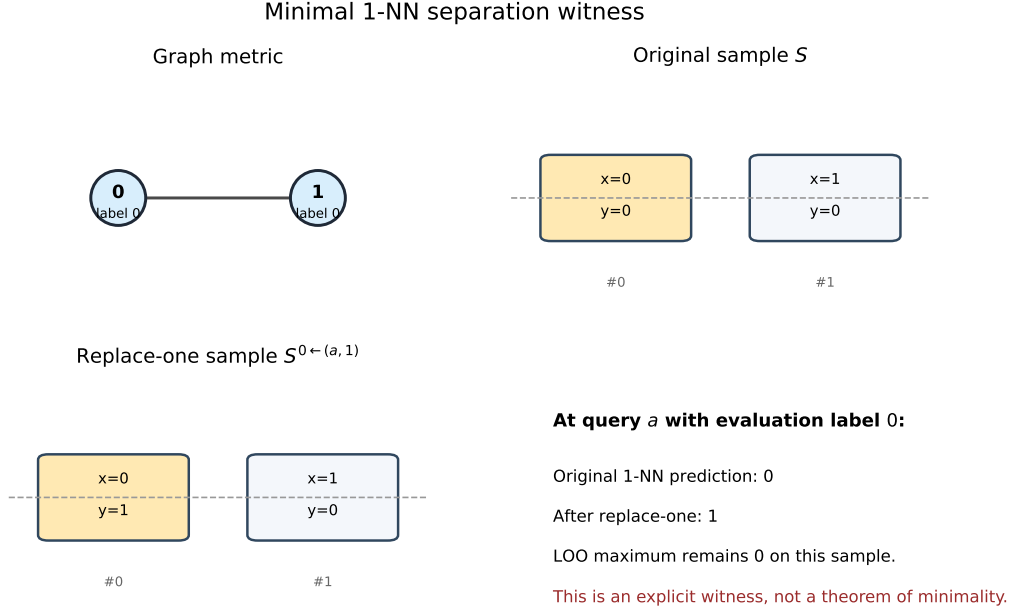


Figure 2: The minimal hand-checkable 1-NN witness. The unperturbed sample is LOO-stable at both occurrences, while a replace-one perturbation changes the prediction at a selected query. Minimality of the two-vertex pattern is currently supported by computational certificates rather than by a formal impossibility proof.

2.1 Algorithmic stability and generalization

Most modern stability results are phrased around replacing one training example and then evaluating expected or fresh-test loss [1, 11, 14]. This perspective is powerful for generalization theory, but it can obscure a simpler definition-level point relevant for local rules: “change one sample point” is not a unique operation. Delete-one, add-one, replace-one, and leave-one-out share a family resemblance, yet they differ both in how the sample changes and in where the learner is evaluated afterward. Our contribution in this section is therefore not a new stability hierarchy in the sense of [1], but a deterministic finite-sample reframing that makes the perturbation operation and the evaluation point explicit.

2.2 Nearest-neighbor classification and deleted estimates

The nearest-neighbor line begins with Fix and Hodges and the classical risk results of Cover and Hart, together with the consistency framework later formalized by Stone [9, 5, 4, 18]. Those works study statistical performance under distributional sampling rather than worst-case single-perturbation effects. Closer to our setting are deleted-estimate and local-rule analyses such as Rogers and Wagner’s finite-sample bound for local discrimination rules and the distribution-free deleted-estimate inequalities of Devroye and Wagner [17, 6, 7]. These are important precedents because they show that leave-one-out style evaluation has a substantial classical history around nearest-neighbor methods. Even so, their goal is not to separate LOO from adversarial replace-one at the level of fixed finite metric witnesses with explicit deterministic tie-breaking.

2.3 Leave-one-out, replace-one, and conformal stability certificates

Kearns and Ron connect algorithmic stability with leave-one-out cross-validation, already emphasizing that a CV-style perturbation has its own logic [10]. Recent conformal-prediction work makes related distinctions operational again, often in service of coverage certificates rather than 0–1 prediction loss. Examples include transductive stability analyses and training-conditional or stable conformal guarantees [15, 13, 16, 12]. We

cite this line not because our paper is a conformal-prediction paper, but because it reinforces the message that leave-one-out style control and replacement-based control can lead to different guarantees. Our setting is more concrete and more local: deterministic k-NN on finite metric spaces with explicit top- k neighbor ordering.

2.4 Finite metric witnesses and local learning rules

Metric-space analyses of k-NN typically appear in consistency or geometric generality results rather than perturbation-separation examples [3]. We did not identify, in the sources inspected for this draft, a prior finite graph-metric example whose explicit purpose is to show that fixed-sample LOO stability need not control fixed-sample adversarial replace-one stability for deterministic 1-NN under the conventions used here. This absence is not enough to justify a sweeping novelty claim. It does, however, support a careful contribution statement: the paper’s distinctive value lies in combining definition hygiene, explicit deterministic k-NN perturbation analysis, and a reproducible finite witness/certificate pipeline.

What this paper does not claim. We do not claim a new generalization bound, a contradiction to classical consistency results, or a theorem-level odd- k lifting result from current computational candidates alone. The paper’s main ambition is narrower and, we think, cleaner: to make precise which one-sample perturbations are being discussed, how they interact at the uniform level, and how sharply they can separate at the pointwise finite-metric level for deterministic nearest neighbors.

3 Definitions and Preliminaries

3.1 Data Model and Sample

Definition 3.1 (Sample). A *sample* S of size n is an ordered tuple

$$S = ((x_1, y_1), \dots, (x_n, y_n))$$

where each $x_i \in X$ lies in a finite metric space (X, d) and each label $y_i \in \{0, 1\}$.

The order of the tuple is semantically relevant only for deterministic tie-breaking. Two identical labeled points appearing at different indices are treated as distinct sample occurrences.

- **Duplicate points are allowed.**
- **Conflicting labels at the same point are allowed.**

Any theorem or experiment that excludes duplicates or conflicts must state this explicitly.

Definition hygiene. The distinctions above are not bookkeeping trivia. Ordered duplicates, conflicting labels, and deterministic tie-breaking are exactly the places where local instability can hide. Several later examples collapse or become ambiguous if these conventions are left implicit.

Definition 3.2 (Finite Metric Space). A finite metric space is a finite set X together with a distance function

$$d : X \times X \rightarrow \mathbb{R}_{\geq 0}$$

satisfying for all $x, x', z \in X$:

1. $d(x, x) = 0$;
2. $d(x, x') > 0$ for $x \neq x'$;
3. $d(x, x') = d(x', x)$;
4. $d(x, z) \leq d(x, x') + d(x', z)$.

Definition 3.3 (Graph Shortest-Path Metric). For a finite, simple, connected, unweighted, undirected graph $G = (V, E)$, the graph metric is

$$d_G(u, v) = \text{length of a shortest path from } u \text{ to } v.$$

Disconnected graphs are rejected in code paths that require a metric.

3.2 Prediction Rule

The query point x may be any element of X , including a point that appears in the training sample.

Note 3.4 (Deterministic Tie-Breaking). Neighbor ordering is determined lexicographically by:

1. smaller distance to the query point;
2. smaller sample index in the ordered tuple.

If class votes tie after the first k ordered neighbors are selected, the predicted label is resolved by fixed label order $0 \prec 1$. Ties are always broken in favor of label 0.

Definition 3.5 (k -NN Prediction). Given $k \in \mathbb{N}$ with $1 \leq k \leq n$, the deterministic k -NN predictor $h_S^{(k)}(x)$ is obtained by:

1. ordering the sample occurrences by the deterministic neighbor rule above;
2. selecting the first k occurrences;
3. taking the majority vote of their labels;
4. resolving label-vote ties by $0 \prec 1$.

3.3 Loss Function

Definition 3.6 (Binary 0-1 Loss). For a hypothesis h and a labeled point $(x, y) \in X \times \{0, 1\}$, the binary classification loss is

$$\ell(h, (x, y)) = \mathbf{1}\{h(x) \neq y\}.$$

3.4 Sample Perturbations

Let $S = ((x_1, y_1), \dots, (x_n, y_n))$.

Definition 3.7 (Delete-One). For index $i \in \{1, \dots, n\}$, define

$$S^{-i} = ((x_1, y_1), \dots, (x_{i-1}, y_{i-1}), (x_{i+1}, y_{i+1}), \dots, (x_n, y_n)).$$

Definition 3.8 (Add-One). For any labeled point $z = (x, y) \in X \times \{0, 1\}$, define

$$S \oplus z$$

as the ordered tuple obtained by appending z at the end.

Definition 3.9 (Replace-One). For index i and labeled point $z = (x', y') \in X \times \{0, 1\}$, define

$$S^{i \leftarrow z}$$

as the ordered tuple obtained by replacing the i -th occurrence of S with z .

In v0.1 worst-case stability definitions, the replacement point ranges over all labeled points in $X \times \{0, 1\}$, not over a data-generating distribution.

3.5 Stability Notions

Unless otherwise stated, all pointwise notions are evaluated with fixed k , fixed metric space, fixed tie-breaking rule, and fixed loss defined above.

Note 3.10. Pointwise stability statements concern a fixed training sample, a fixed perturbation, and a fixed evaluation pair (x, y) . Expected stability refers to the expectation under an explicitly specified data-generating distribution. Uniform stability is the supremum of the corresponding indicator over all valid ordered training tuples, all allowed perturbation choices, and all evaluation pairs.

Definition 3.11 (Pointwise Delete-One Stability Indicator). For index i and query-label pair (x, y) , define

$$\Delta_{\text{del}}(S, i, x, y) = \left| \ell(h_S^{(k)}, (x, y)) - \ell(h_{S^{-i}}^{(k')}, (x, y)) \right|,$$

where $k' = \min(k, n - 1)$ when deletion reduces sample size below k .

Definition 3.12 (Pointwise Replace-One Stability Indicator). For index i , replacement z , and query-label pair (x, y) , define

$$\Delta_{\text{rep}}(S, i, z, x, y) = \left| \ell(h_S^{(k)}, (x, y)) - \ell(h_{S^{i \leftarrow z}}^{(k)}, (x, y)) \right|.$$

Definition 3.13 (Pointwise Add-One Stability Indicator). For added labeled point z and query-label pair (x, y) , define

$$\Delta_{\text{add}}(S, z, x, y) = \left| \ell(h_S^{(k)}, (x, y)) - \ell(h_{S \oplus z}^{(k)}, (x, y)) \right|.$$

Definition 3.14 (LOO / CVloo Stability). For each index i , define

$$\Delta_{\text{loo}}(S, i) = \left| \ell(h_{S^{-i}}^{(k')}, (x_i, y_i)) - \ell(h_S^{(k)}, (x_i, y_i)) \right|.$$

LOO evaluates at the deleted sample occurrence itself. This definition is occurrence-based: if the same point appears twice, deleting one copy and evaluating at that occurrence is distinct from deleting the other.

For LOO, the uniform stability supremum is over ordered tuples S and indices i , with the evaluation pair fixed to the deleted occurrence.

3.6 Consistency Note

The consistency notion used in paper-level discussion is classical distributional consistency for k -NN classification. It is not a finite-sequence surrogate notion. Any finite-metric experiment provides evidence about worst-case stability behavior; it does not by itself constitute a consistency statement.

4 Uniform Stability Calculus

At the level of perturbation operations, replace-one is the composite move obtained by deleting an occurrence and then inserting a new labeled point. That observation is almost tautological at the sample level, but it is worth writing down explicitly because it yields the correct direction of implication for worst-case stability control.

Proposition 4.1 (Uniform replace-one through delete-plus-add). *Fix a deterministic learning rule A that maps ordered samples to predictors and a bounded loss $\ell \in [0, 1]$. Suppose that for every sample S , every index i , every evaluation pair (x, y) , and every admissible appended labeled point z ,*

$$\left| \ell(A(S), (x, y)) - \ell(A(S^{-i}), (x, y)) \right| \leq \beta_{\text{del}}$$

and

$$\left| \ell(A(T), (x, y)) - \ell(A(T \oplus z), (x, y)) \right| \leq \beta_{\text{add}}$$

uniformly over all intermediate samples T . Then every replace-one perturbation satisfies

$$\left| \ell(A(S), (x, y)) - \ell(A(S^{i \leftarrow z}), (x, y)) \right| \leq \beta_{\text{del}} + \beta_{\text{add}}.$$

uniform bound
~~replace-one~~ — ~~delete-one~~ + add-one

LOO ~~delete-one~~ at deleted occurrence

Uniform relations do not imply that pointwise LOO controls pointwise replace-one.

Figure 3: Operation-level factorization of a replace-one perturbation into a delete-one step followed by an add-one step. The proof of [Theorem 4.1](#) is just the triangle inequality on the corresponding binary losses.

Proof. For fixed $S, i, z, (x, y)$, let $T = S^{-i}$. By construction, $S^{i \leftarrow z} = T \oplus z$ up to the repository's fixed convention that the new occurrence is inserted after the deletion. Therefore

$$\begin{aligned} |\ell(A(S), (x, y)) - \ell(A(S^{i \leftarrow z}), (x, y))| &\leq |\ell(A(S), (x, y)) - \ell(A(T), (x, y))| \\ &\quad + |\ell(A(T), (x, y)) - \ell(A(T \oplus z), (x, y))| \end{aligned}$$

by the triangle inequality, and the claimed bound follows from the two uniform assumptions. $\square$

[Theorem 4.1](#) is deliberately abstract. It is not specific to k -NN, and it does not yet tell us whether the bound is informative for a particular local learner. Its role is to freeze the algebra of perturbation types before we zoom in on the exact combinatorics of deterministic nearest neighbors.

The proposition also clarifies what it does *not* say. A pointwise LOO indicator cannot simply be dropped into the role of delete-one control above, because LOO changes the evaluation point along with the sample. Similarly, a fresh-test or expected-loss stability statement from the classical literature does not automatically transfer to the worst-case fixed-query indicators used in the present paper. The whole point of the calculus is to prevent these category mistakes.

5 Exact Deterministic k -NN Perturbation Criterion

Once the perturbation type is fixed, a deterministic k -NN prediction can only change for one structural reason: the signed class vote inside the ordered top- k neighborhood crosses zero. This section makes that statement exact.

For an ordered sample S and query x , let $\pi_S^x(1), \dots, \pi_S^x(n)$ be the sample indices sorted by increasing distance to x and then by sample index. Write

$$N_k(S, x) = (\pi_S^x(1), \dots, \pi_S^x(k))$$

for the ordered top- k neighbor list and define the signed vote margin

$$M_k(S, x) = \sum_{r=1}^k (2y_{\pi_S^x(r)} - 1).$$

Under the label-order tie-break $0 \prec 1$, the deterministic classifier predicts label 1 exactly when $M_k(S, x) > 0$.

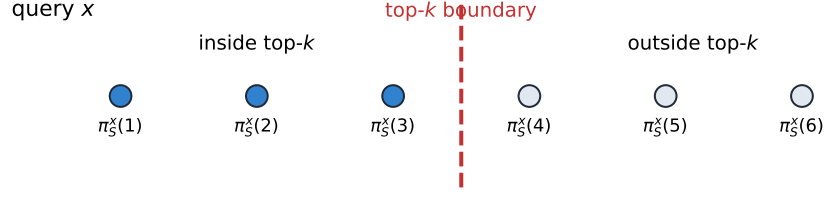


Figure 4: Ordered top- k membership determines the relevant local state for deterministic k -NN. Instability is possible only when a perturbation changes this neighborhood or flips the labels of occurrences already inside it.

Outside-top- k deletion leaves the top- k vote unchanged



Figure 5: Deletion outside the ordered top- k neighborhood is inert unless it changes the boundary and promotes a new occurrence into the vote.

Proposition 5.1 (Margin crossing criterion). *Fix S , k , and a query x . Let S' be any perturbed sample of the same ambient metric space, interpreted with the same deterministic tie-breaking rule. Then*

$$h_S^{(k)}(x) \neq h_{S'}^{(k)}(x)$$

if and only if the signed vote margins satisfy

$$(M_k(S, x) \leq 0 < M_k(S', x)) \quad \text{or} \quad (M_k(S', x) \leq 0 < M_k(S, x)).$$

Proof. By definition of the deterministic vote rule, the prediction is 1 exactly for strictly positive signed margin and 0 otherwise. Therefore the prediction changes precisely when one margin is positive and the other is non-positive. $\square$

Theorem 5.1 is elementary but useful: every perturbation argument can be reduced to tracking how the ordered top- k membership changes and how the corresponding signed vote changes.

Two immediate consequences are worth isolating. First, deleting an occurrence outside $N_k(S, x)$ cannot change the prediction at x unless the deletion also shifts the boundary and pulls a new occurrence into the top- k set. Second, a replace-one perturbation can act in two different ways: it can change the geometry of who enters the neighborhood, or it can leave the geometry fixed and still flip the vote by changing the label attached to a strategically placed occurrence.

These mechanisms become especially sensitive in the presence of duplicate occurrences and equal-distance contenders, because the top- k neighborhood is then controlled by the deterministic tie-breaking convention rather than by distance alone. That is not a nuisance artifact. It is part of the object being studied.

Removing an in-top-k occurrence can expose a new boundary neighbor

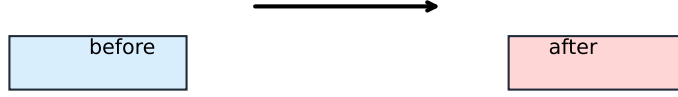


Figure 6: A local perturbation changes the prediction only when the top- k signed vote crosses zero. This is the exact mechanism behind all explicit witnesses in the draft.

Replace-one can swap label mass across the top-k threshold



Figure 7: Replace-one perturbations can flip the decision either by moving a new occurrence into the decisive neighborhood or by changing the label composition of a neighborhood that was already geometrically close to the boundary.

The rest of the paper repeatedly uses this margin-crossing microscope. In the 1-NN witness of [Section 6](#), the decisive effect is a single nearest-occurrence replacement. In the odd- k gadget search of [Section 7](#), the search heuristics attempt to engineer repeated occurrence patterns whose signed margin is stable under LOO but fragile under a targeted replace-one move.

6 Finite Metric Separation for 1-NN

We now come to the main theorem-level witness of the draft. The statement is small, explicit, and completely deterministic.

Proposition 6.1 (Pointwise LOO does not control pointwise replace-one for deterministic 1-NN). *There exists a finite graph metric space, an ordered labeled sample S , and a query-label pair (x, y) such that*

$$\Delta_{\text{loo}}(S, i) = 0 \quad \text{for every sample index } i,$$

while for some index j and some admissible replacement point z ,

$$\Delta_{\text{rep}}(S, j, z, x, y) = 1.$$

Proof. Let the metric space contain two vertices a, b joined by one edge, so $d(a, b) = 1$. Consider the ordered sample

$$S = ((a, 0), (b, 0)).$$

Use deterministic 1-NN with the fixed tie-breaking conventions from [Section 3](#).

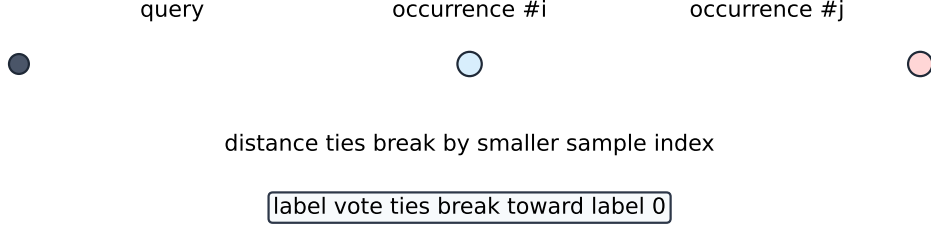


Figure 8: Tie-breaking is not decorative. When several occurrences are at equal distance, the deterministic sample order and fixed label order can determine whether a perturbation is visible to the predictor.

For LOO, delete the first occurrence. The remaining sample is $((b, 0))$, whose 1-NN prediction at the deleted occurrence’s location a is still 0 because the sole remaining neighbor has label 0. Hence the loss at $(a, 0)$ is unchanged. The same argument applies when deleting the second occurrence and evaluating at $(b, 0)$. Therefore $\Delta_{\text{loo}}(S, 1) = \Delta_{\text{loo}}(S, 2) = 0$.

Now evaluate at the fixed query-label pair $(x, y) = (a, 0)$. Replace the first occurrence $(a, 0)$ by $(a, 1)$, obtaining

$$S^{1 \leftarrow (a, 1)} = ((a, 1), (b, 0)).$$

At query a , the original sample predicts 0 because $(a, 0)$ is the unique nearest occurrence. In the perturbed sample, the unique nearest occurrence is now $(a, 1)$, so the prediction becomes 1. Hence $\Delta_{\text{rep}}(S, 1, (a, 1), a, 0) = 1$. $\square$

Figure 2 gives the same witness visually. What matters is not that the example is exotic, but that it is minimal enough to inspect line-by-line. The phenomenon already appears before any asymptotics, averages, or large combinatorial constructions enter the story.

The repository also contains a finite-search pipeline that looks for smaller or structurally simpler witnesses under various restrictions. The current output is best interpreted as follows. The witness in Theorem 6.1 is theorem-level because it is explicit and hand-checkable. Claims that no witness exists below a certain search frontier are *not* theorem-level; they are computational certificates tied to the enumerated search space.

This distinction matters. A paper can responsibly say “here is a minimal explicit witness we know” and separately say “here is a reproducible finite certificate supporting minimality within the searched regime.” It should not blur those two statements together. That boundary is part of the contribution, not an embarrassment to hide.

7 General k and Gadget Candidates

The 1-NN witness isolates the conceptual point cleanly, but it is natural to ask whether the same separation pattern persists for larger odd values of k . Our current repository contains a bounded finite search for odd- k gadget candidates, together with plots and machine-readable output. What it does not yet contain is a fully formal lifting theorem. This section is written to make that status transparent.

The search heuristic is motivated by a repeated-occurrence amplification idea. If a 1-NN witness works because one decisive nearest occurrence can be replaced, then for odd k one may hope to replicate that decisive role across several tightly controlled occurrences so that LOO remains benign while a targeted replace-one move still crosses the top- k signed margin. The search pipeline explores deterministic finite samples designed around this pattern.

The computational record suggests the following formulation.

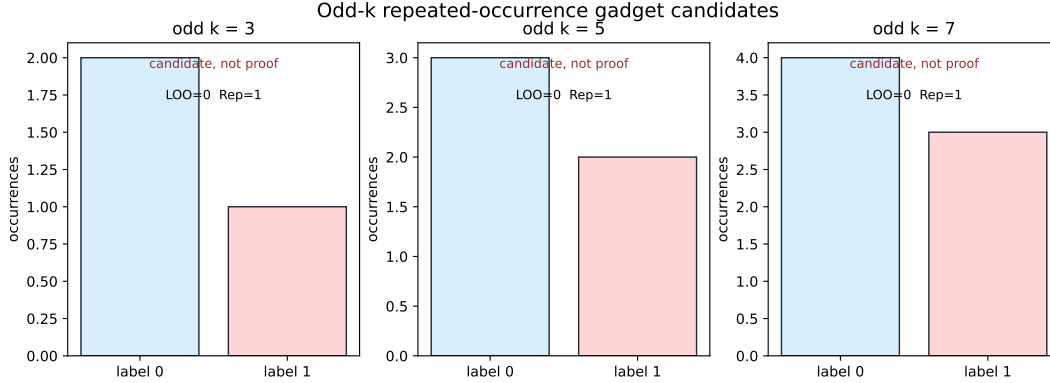


Figure 9: Representative computational candidates for odd- k separations. These outputs are evidence for a plausible lifting pattern, not theorem statements.

Conjecture 7.1 (Odd- k repeated-occurrence lifting). *For every odd $k \geq 1$, there exists a deterministic finite metric sample for k -NN such that pointwise LOO stability holds at the designated occurrences while pointwise replace-one instability occurs at a chosen query-label pair.*

Why not state this as a theorem? Because the present evidence is still a search result plus a design heuristic. The candidate families for $k = 3, 5, 7$ are real and reproducible, but the repository does not yet contain a general proof that the construction scales to all odd k , nor a proof that the searched examples capture the true minimal combinatorics.

That said, the candidates are still scientifically useful. First, they test the robustness of the mechanism seen in [Theorem 6.1](#): the separation is not obviously a one-neighbor anomaly. Second, they tell us which features seem structurally important for any future proof, namely repeated local blocks, controlled vote margins, and deterministic tie-sensitive neighbor ordering. Third, they warn us that an honest paper should keep a clear gradient of claim strength: theorem, computational certificate, conjecture.

Even- k appears subtler. Because the vote boundary already sits at a tied margin, deterministic label-order tie-breaking plays a larger role, and the search space is less cleanly explained by odd-margin amplification. We therefore make no even- k theorem claim and no impossibility claim here.

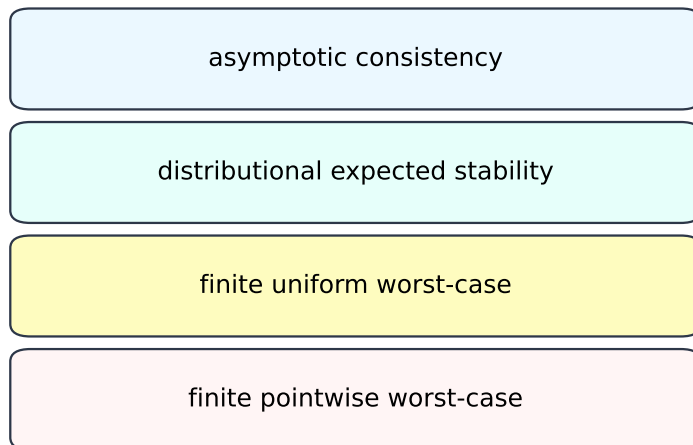
8 Consistency Compatibility

It is worth pausing to say what these finite worst-case separations do *not* imply. Classical nearest-neighbor theory studies risk under a distributional sampling model and asks whether the learned classifier converges to the Bayes rule or to a near-optimal risk level as sample size grows [\[5, 4, 18\]](#). Our results operate on a different axis: fixed finite samples, deterministic tie-breaking, and worst-case one-perturbation sensitivity.

There is therefore no contradiction between the statements “deterministic finite 1-NN can be replace-one unstable at a fixed query” and “ k -NN is consistent under the usual asymptotic hypotheses.” The former is a local combinatorial fact about a chosen finite configuration. The latter is a distributional limit theorem. Both can be true simultaneously, and in the current draft they are.

This distinction also helps orient the relation to conformal prediction and LOO-style calibration analyses. In those literatures, one often studies how deleting or exchanging points affects a prediction set, a conformity score, or a coverage certificate under exchangeability or training-conditional assumptions. Those are important questions, but they are not identical to the present 0–1-loss perturbation calculus for deterministic k -NN. Our witness results should therefore be read as a mechanism-level complement to that literature, not as a replacement for it.

The practical lesson is simple. If one wants a stability notion for a local learner, one should first decide whether the goal is generalization control, cross-validation behavior, adversarial local robustness, or training-conditional prediction stability. The same phrase “change one sample point” cannot safely do all four jobs at once.



Sections 6 and 7 live near the top; Section 8 explains why the bottom layer is different.

Figure 10: Finite worst-case perturbation sensitivity, classical asymptotic consistency, and conformal-style conditional validity live in different analytical regimes. The present paper sits in the first regime.

9 Discussion

The central payoff of this draft is conceptual clarity with enough concrete mathematics behind it to be checkable. The paper does not merely rename familiar notions. It shows that once perturbation type, evaluation point, and deterministic tie-breaking are all fixed explicitly, one gets both a clean uniform perturbation calculus and a sharp pointwise separation phenomenon for nearest neighbors.

At the current stage, the strongest established claims are the following. Replace-one admits a delete-plus-add upper bound at the uniform loss level ([Theorem 4.1](#)). Deterministic k -NN prediction changes are exactly margin-crossing events inside the ordered top- k neighborhood ([Theorem 5.1](#)). And deterministic 1-NN already exhibits a finite metric separation between pointwise LOO stability and pointwise replace-one stability ([Theorem 6.1](#)).

The weaker parts of the story are equally important to state honestly. The minimality of the current 1-NN witness beyond the explicit construction is supported by computational certificates, not by a full impossibility proof. The odd- k extension is presently a conjectural lifting pattern backed by bounded search evidence for $k = 3, 5, 7$. Those are meaningful research assets, but they should not be promoted beyond their evidentiary level.

The draft also has several limitations that shape the next phase.

- (1) The current theorem-level separation is for deterministic finite 1-NN; the general odd- k story still lacks a formal proof.
- (2) The ambient spaces are finite metric spaces, often realized as graph shortest-path metrics. This gives crisp witnesses but does not yet address broader geometric classes.
- (3) The paper studies binary 0–1 loss. Real-valued scores, multiclass labels, and regression analogues remain open.
- (4) The discussion is worst-case and fixed-sample. It is intentionally not a statistical rate paper.
- (5) The computational certificates are reproducible and well documented, but they are still certificates rather than formal proofs.

These limitations point directly to the most natural follow-up problems.

- (1) Prove or refute [Theorem 7.1](#).
- (2) Strengthen the current certificate pipeline into a formal minimality argument or an automated proof assistant artifact.
- (3) Characterize which metric-space constraints eliminate the separation.
- (4) Extend the perturbation calculus to other local rules, such as weighted neighbors or radius-based classifiers.
- (5) Connect the finite worst-case indicators here to distributional stability notions without collapsing the distinctions this paper isolates.

If there is a broader moral, it is this: for local learning rules, the geometry of the neighborhood and the semantics of the perturbation must be specified together. Once that is done, some relationships become cleaner than expected, and some purportedly similar notions separate much sooner than intuition suggests.

A Proof Sketches and Deferred Proofs

This appendix records the short proofs whose main ideas already appear in the body, together with the claim-strength boundaries that matter for review.

A.1 Uniform perturbation calculus

The proof of [Theorem 4.1](#) is intentionally elementary. Its value is not technical depth but category control: replace-one is an operation composition, whereas LOO is an operation-plus-evaluation convention. The triangle-inequality argument therefore applies directly to delete/add versus replace, but not to LOO without additional translation lemmas.

A.2 Deterministic k -NN perturbation criterion

The proof of [Theorem 5.1](#) reduces the classifier to a signed vote threshold. All geometric complexity is compressed into the ordered top- k neighbor list $N_k(S, x)$. Once that list is fixed, the classifier is just the sign of $M_k(S, x)$ with the non-positive side resolved toward label 0. The only way a perturbation changes the prediction is therefore to move the signed margin across zero.

A.3 Two-point 1-NN witness

The witness of [Theorem 6.1](#) is fully explicit and hand-checkable. No exhaustive search is needed for the existence claim itself. What the repository’s search artifacts contribute is different: they show that within the enumerated search spaces, no one-vertex witness appears and the two-vertex pattern is the first observed regime where the separation occurs. That latter statement remains certificate-level evidence, not a proof.

B Enumeration Protocol

This appendix summarizes the finite search spaces used to produce the witness and certificate artifacts cited in the main text. The guiding rule throughout the repository is that finite enumeration can support reproducible evidence, but it does not upgrade a claim to theorem status unless paired with a formal argument.

B.1 1-NN witness search

The baseline search enumerates connected simple undirected graphs up to the configured vertex cutoff, all binary labelings, and all sample orderings under the deterministic conventions of [Section 3](#). For each candidate sample it computes the fixed-sample maxima of the LOO and replace-one indicators and records witnesses where the former is 0 and the latter is 1.

B.2 Tie-free filtering

The tie-free post-processing stage rechecks accepted witness records and keeps only those whose decisive predictions involve a unique nearest sample occurrence in the original and perturbed configurations. In the current artifact set, the minimal two-vertex witness survives this filter.

B.3 Odd- k gadget candidate search

The odd- k search uses bounded deterministic constructions designed around repeated-occurrence margin amplification. The output is summarized in Table 5 and is explicitly marked as computational evidence only.

C Additional Figures

The main text uses only the figures that directly support the narrative spine of the paper. The repository contains a somewhat richer figure set under ‘outputs/figures/’, including intermediate mechanism schematics and witness renderings used during drafting. These assets are part of the reproducibility record and can be promoted into the main text later if venue constraints allow.

D Bibliographic Notes

The bibliography is curated to support three distinct jobs. First, it anchors the classical algorithmic-stability vocabulary [1, 11, 8, 2]. Second, it situates nearest-neighbor performance and deleted-estimate analyses [5, 4, 18, 17, 6, 7]. Third, it records the recent prediction-stability and conformal-certification papers that motivate keeping LOO-style and replacement-style perturbations distinct [15, 13, 16, 12].

The novelty assessment and claim registry in ‘docs/’ remain the authoritative location for broader scope decisions. The manuscript bibliography is therefore best read as the paper-facing subset of that larger internal review record.

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Work	Perturbation	Evaluation point	Guarantee type	k-NN?	Relation to this paper
Fix and Hodges, *Discriminatory Analysis*	not a stability paper	prediction point under classification rule	distributional / classical statistical	no	historical origin of NN line, useful context only
Cover and Hart, *Nearest Neighbor Pattern Classification*	not a perturbation paper	prediction risk under i.i.d. sampling	distributional / asymptotic	yes	important to separate our finite worst-case focus from classical consistency lore
Cover, *Estimation by the Nearest Neighbor Rule*	not a perturbation paper	prediction under sampling distribution	distributional / asymptotic	yes	supports consistency-compatibility discussion
Stone, *Consistent Nonparametric Regression*	not a perturbation paper	population risk	distributional / asymptotic	no	key citation for "finite instability does not contradict consistency"
Rogers and Wagner, *A Finite Sample Distribution-Free Performance Bound for Local Discrimination Rules*	delete-one style deleted estimate	deleted observation / cross-validation style	mostly distribution-free finite-sample	no	nearest known classical bridge between local rules and delete-one evaluation
Devroye and Wagner, *Distribution-Free Inequalities for the Deleted and Holdout Error Estimates*	delete-one / holdout	deleted point or held-out point	distribution-free finite-sample	no	central precedent for leave-one-out style evaluation, but not our worst-case perturbation calculus
Devroye and Wagner, *The Strong Uniform Consistency of Nearest Neighbor Density Estimates*	not a stability paper	population estimation	asymptotic	yes	background only
Kearns and Ron, *Algorithmic Stability and Sanity-Check Bounds for Leave-One-Out Cross-Validation*	leave-one-out and general stability notions	deleted observation / CV	statistical / generalization-oriented	no	strong precedent that LOO and stability interact, but not a k-NN finite-metric separation paper

Table 1: Condensed map of representative references relative to the present deterministic finite-metric perspective.

This map is a condensed companion to docs/literature/LIT_TABLE.md.

Notion	Perturbation	Evaluated at	Supremum over	Distribution?	Used for
delete-one	remove one occurrence	arbitrary query-label pair	indices and queries	no	uniform/pointwise perturbation analysis
replace-one	swap one occurrence for adversarial labeled point	arbitrary query-label pair	indices, replacements, queries	no	strong local perturbation control
add-one	append one occurrence	arbitrary query-label pair	added points and queries	no	decomposition of replace-one
LOO	delete one occurrence	that deleted occurrence	delete index only	no	cross-validation style local stability
uniform stability	usually replace-one in classical literature	fresh test point or worst-case query	all samples	often yes	generalization bounds

Table 2: Comparison of perturbation notions used in this paper and in the surrounding stability literature. Computational witness sections use fixed-sample worst-case indicators, not distributional expected stability.

Vertices	Edges	Sample	k	max LOO	max replace-one	Status
2	1	order=[0, 1]; labels=[0, 0]	1	0	1	explicit witness; certificate-level minimality only
2	1	order=[0, 1]; labels=[1, 1]	1	0	1	explicit witness; certificate-level minimality only
2	1	order=[1, 0]; labels=[0, 0]	1	0	1	explicit witness; certificate-level minimality only
2	1	order=[1, 0]; labels=[1, 1]	1	0	1	explicit witness; certificate-level minimality only

Table 3: Minimal explicit witness patterns for the 1-NN separation story under deterministic tie-breaking. The witness row is hand-checkable, while minimality over the search space remains computational evidence.

Artifact	Observed min vertices	No-solution range	Hash	Status
TASK-007	2	[1]	5b7a8925688a...	computational evidence only
TASK-008	2	[1]	a52aa8a77a9a...	computational evidence only

Table 4: Summary of the current computational minimality certificates for the 1-NN witness search.
Certificate tables summarize finite search outputs and do not upgrade them to proof.

k	vertices	sample size	max LOO	max replace-one	status	script
3	2	3	0	1	computational candidate only	experiments/search_k_gadgets.py
5	2	5	0	1	computational candidate only	experiments/search_k_gadgets.py
7	2	7	0	1	computational candidate only	experiments/search_k_gadgets.py

Table 5: Representative odd-k gadget candidates returned by the finite search pipeline.
Odd-k rows summarize one representative candidate per k and are not theorem statements.

Artifact	Script	Command	Output hash	Status
tests	python -m pytest	E:\{\}anaconda3\{\}envs\clean\{\}python.exe -m pytest	4D1D10- DUCIBILITY.md	passed
minimal 1-NN search	search_minimal_1nn.py	E:\{\}anaconda3\{\}envs\clean\{\}python.exe experiments\{\}search_minimal_1nn.py -max_vertices 4	5D7A802568A...	generated
tie-free search	search_tie_free.py	E:\{\}anaconda3\{\}envs\clean\{\}python.exe experiments\{\}search_tie_free.py -input out-puts\{\}witnesses\{\}1nn_separation_witnesses.json -output out-puts\{\}witnesses\{\}1nn_tie_free_witnesses.json	52A48A177A9A...	generated
minimality certificate	certify_minimality.py	E:\{\}anaconda3\{\}envs\clean\{\}python.exe experiments\{\}certify_minimality.py -separation out-puts\{\}witnesses\{\}1nn_separation_witnesses.json -tie_free out-puts\{\}witnesses\{\}1nn_tie_free_witnesses.json -json_output out-puts\{\}witnesses\{\}1nn_minimality_certificate.json -markdown_output out-puts\{\}tables\{\}1nn_minimality_certificate.md	92F85B16F79D...	generated
k-gadget search	search_k_gadgets.py	E:\{\}anaconda3\{\}envs\clean\{\}python.exe experiments\{\}search_k_gadgets.py -k_values 3 5 7 -min_vertices 2 -max_vertices 2 -length_modes equal_k - max_candidates_per_k 3	B3204D09A02A...	generated

Table 6: Compact reproducibility map for the current code and artifact pipeline.
Full commands and environment details are listed in outputs/REPRODUCIBILITY.md.